

ARAB YOUTH SPACE HACKATHON 2026 • CHALLENGE 813 • THEME 2

QAYDH القيقظ

Urban heat risk, through the eyes of satellites

Team T0049 • United Arab Emirates

AGENDA

From where it is hot to where we act first

1. Team
2. The heat problem
3. QAYDH
4. Approach & tools
5. The Musaffah story
6. Proof
7. Impact & value
8. Closing

TEAM T0049

Three people, one city problem

انتصار الحبسي

Team lead

نورة الهاجري

Team member

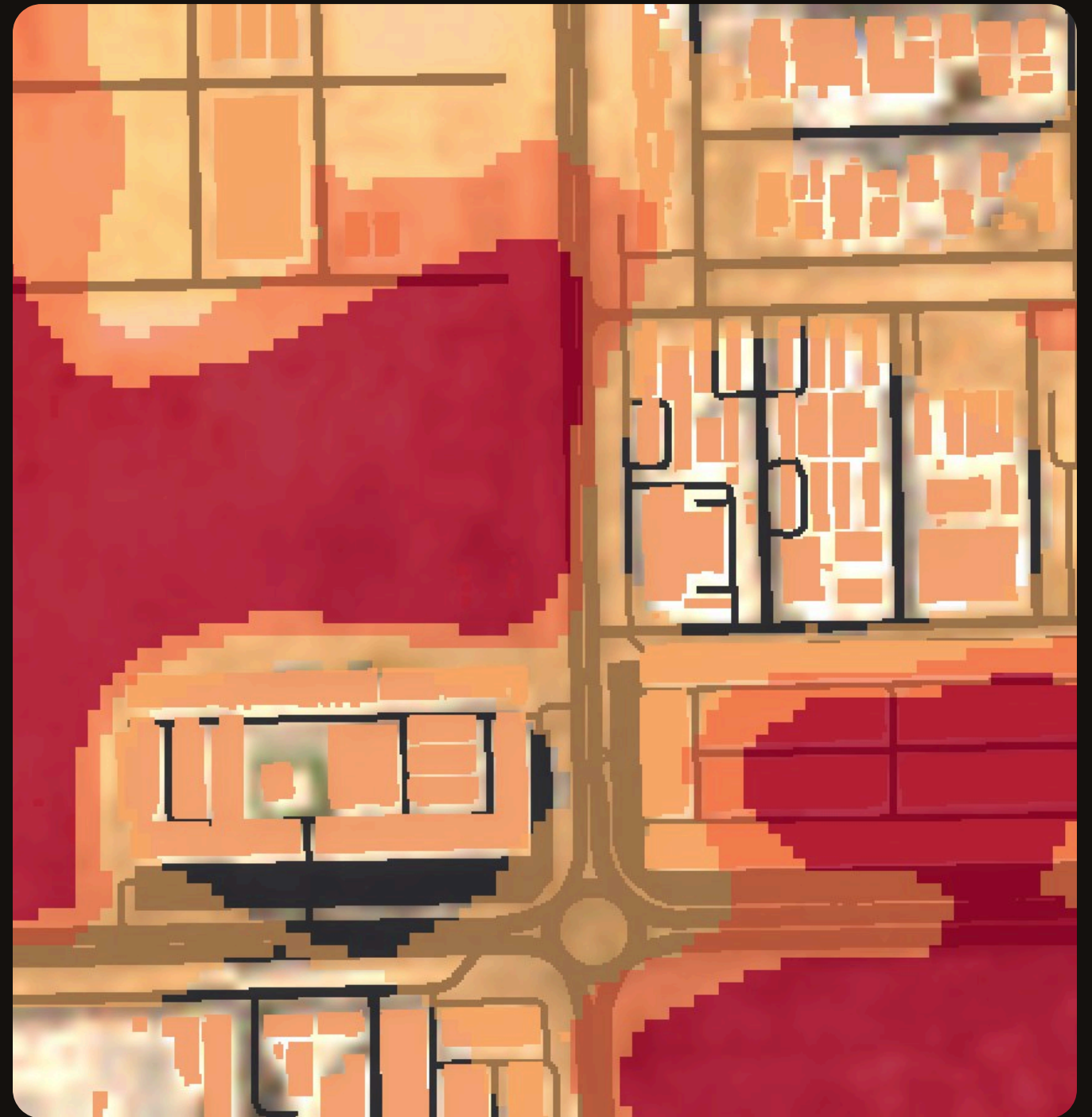
مريم البني

Team member

QAYDH • القيث

The fierce heat of summer, mapped where people live and work

Heat • surfaces • people • action, from open
satellite data.



Why QAYDH as a solution?

What is its impact?

CHALLENGE

Surfaces pass 55 °C. Cooling budgets are spent case by case.

SIGNIFICANCE

Workers, riders, worshippers and bus riders are outside 11:00–20:00.

IMPACT

Shade, cool surfaces and trees go first where people are exposed.

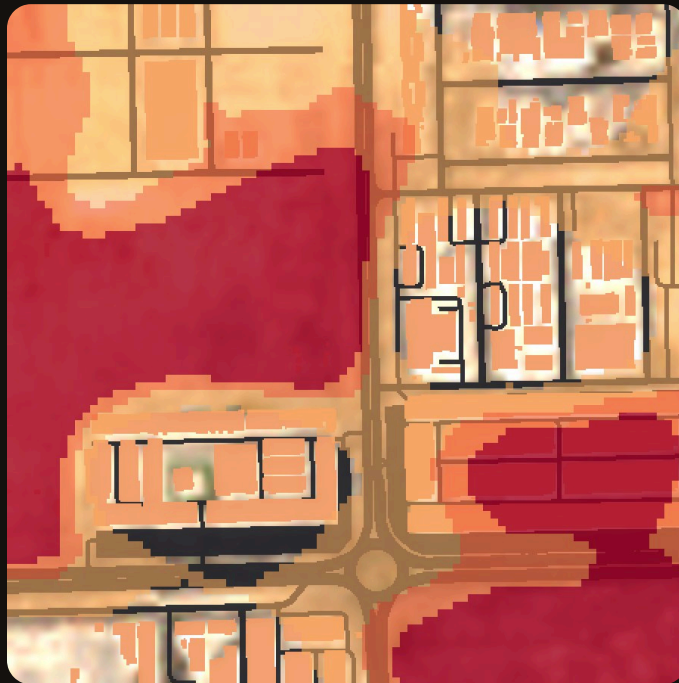
SOLUTION

Thermal + hyperspectral + 10 m surfaces + OSM → one ranked action map.



MUSAFFAH, ABU DHABI

Real places, not pixels



Hotspot M-001

P93 heat · 54% pavement



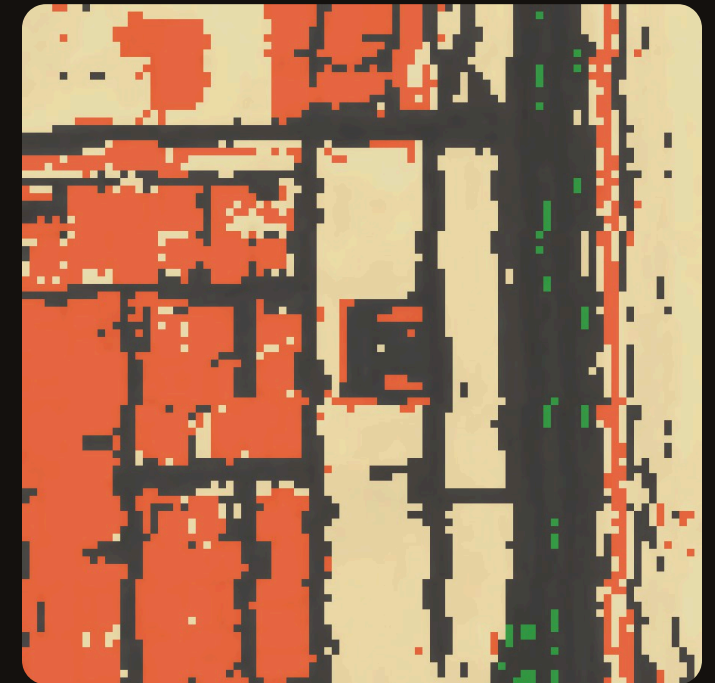
Bus stop · (unnamed)

P97 heat · 0.0% green



Mosque · Mosque inside camp

P75 heat · nearest green 306 m



What is physically there

road · roof · sand · green at 10 m

Solution Overview

1

Where is heat high?

Landsat hazard zones, 3-summer persistence, danger hours

2

Who may be exposed?

Bus stops, mosques, clinics, camps, routes + WorldPop

3

What is there?

Road · roof · sand · green · water at 10 m; materials from Tanager

4

Why may it be hot?

Spatially validated driver model

5

What should be done?

Ranked action, reason, relative potential, alerts

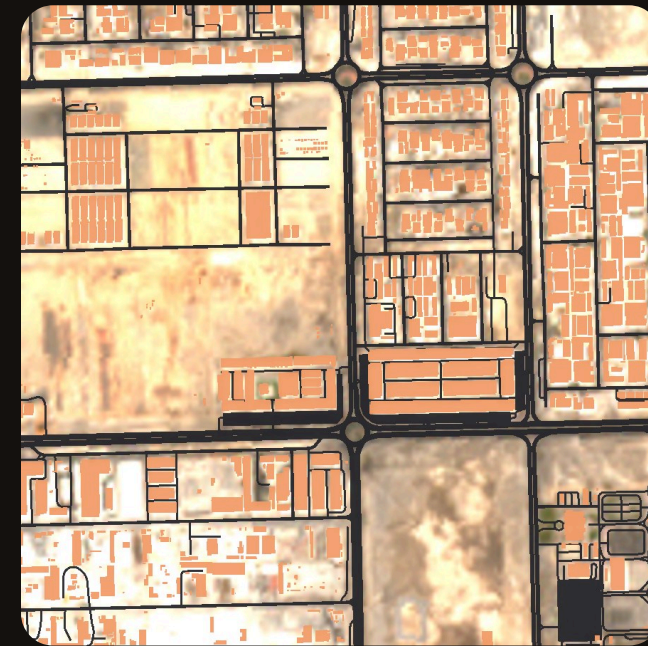
QAYDH القِيظ



Heat



Surfaces



People & places

Data & tools

DATA

Landsat 8/9 thermal · Sentinel-2 10 m
· Planet Tanager hyperspectral · ESA
WorldCover · OSM · Microsoft
footprints · WorldPop · ERA5

TOOLS

Python · Planetary Computer · scikit-
learn · QGIS-ready GeoJSON · gIQ-
ready dashboard

STEPS

Masks → annotation → training →
spatial validation → priority →
dashboard

QAYDH Approach

NOW · POC

Musaffah at 10 m, end to end ·
hyperspectral proof on Riyadh
(Tanager) · Abu Dhabi screening

NEXT · MVP

Satellite 813 / MBZ-SAT over Abu
Dhabi, Dubai, Al Ain · field-checked
labels · gIQ

LATER

Every Gulf city, every summer: one
brief, one map, one budget line

Hi-res imagery

Satellite

Buildings & roads

Heat hazard

Surfaces

Confidence

People places

Exposure sites

Priority

01 · WHERE IS HEAT HIGH?

Hazard zones, not fake street temperatures

Landsat thermal (≈ 100 m) shows zones: elevated ≥ 56.0 °C, high ≥ 57.0 °C, extreme ≥ 57.5 °C. It never pretends to know one bus stop's exact temperature.

57.5 °C

extreme threshold (P95)

55.0 °C

block median

02 · WHO MAY BE EXPOSED?

Bus stops, mosques, clinics, labour camps

139 named outdoor sites from OpenStreetMap, each scored within 150 m: heat percentile, vegetation, impervious cover, distance to green.

1

very-high sites

7

high sites

03 · WHAT IS PHYSICALLY THERE?

Road, roof, sand, plants, water at 10 m

Expert annotation: GIS candidates kept only at $\geq 80\%$ purity, mixed pixels excluded, noisy labels removed, scored on held-out 1 km blocks.

0.91

macro-F1, held-out blocks

0.47

starter index rules

04 · WHY MAY IT BE HOT?

Asphalt and sand up. green down



01 · WHERE IS HEAT HIGH?

Hazard zones, not fake street temperatures

Extreme ≥ 57.5 °C. Landsat gives the zone; buildings and places tell us what falls inside it.

Heat hazard zones (Landsat only)
■ Elevated ≥ 56.0 °C ■ High ≥ 57.0 °C

■ Extreme ≥ 57.5 °C

■ Buildings ■ Roads

Musaffah deep dive

Abu Dhabi · 9 × 9 km at 10 m

WHERE	Persistent summer hotspots at 300 m
WHEN	Hours above 40 °C, from weather reanalysis
WHY	Roof, road, sand or missing greenery, read from hyperspectral spectra
WHO	Residents + people outdoors: mosque-goers, bus riders, delivery riders, workers, children
WHAT NEXT	One named action per hotspot, with estimated °C and confidence

Open the last chapter, “Where to act first”, and pick a hotspot.

Open data: Planet Tanager (CC-BY-4.0), Landsat 8/9, ESA WorldCover, WorldPop 2025, OpenStreetMap, Microsoft building footprints, Open-Meteo/ERA5

Hi-res imagery

Satellite

Buildings & roads

Heat hazard

Surfaces

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People places

Exposure sites

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starter index rules

04 · WHY MAY IT BE HOT?

Asphalt and sand up, green down

A spatially cross-validated model links each 100 m cell's surface mix to its heat. Drivers are shown as associations, not proof of cause.

0.85

R², spatial CV

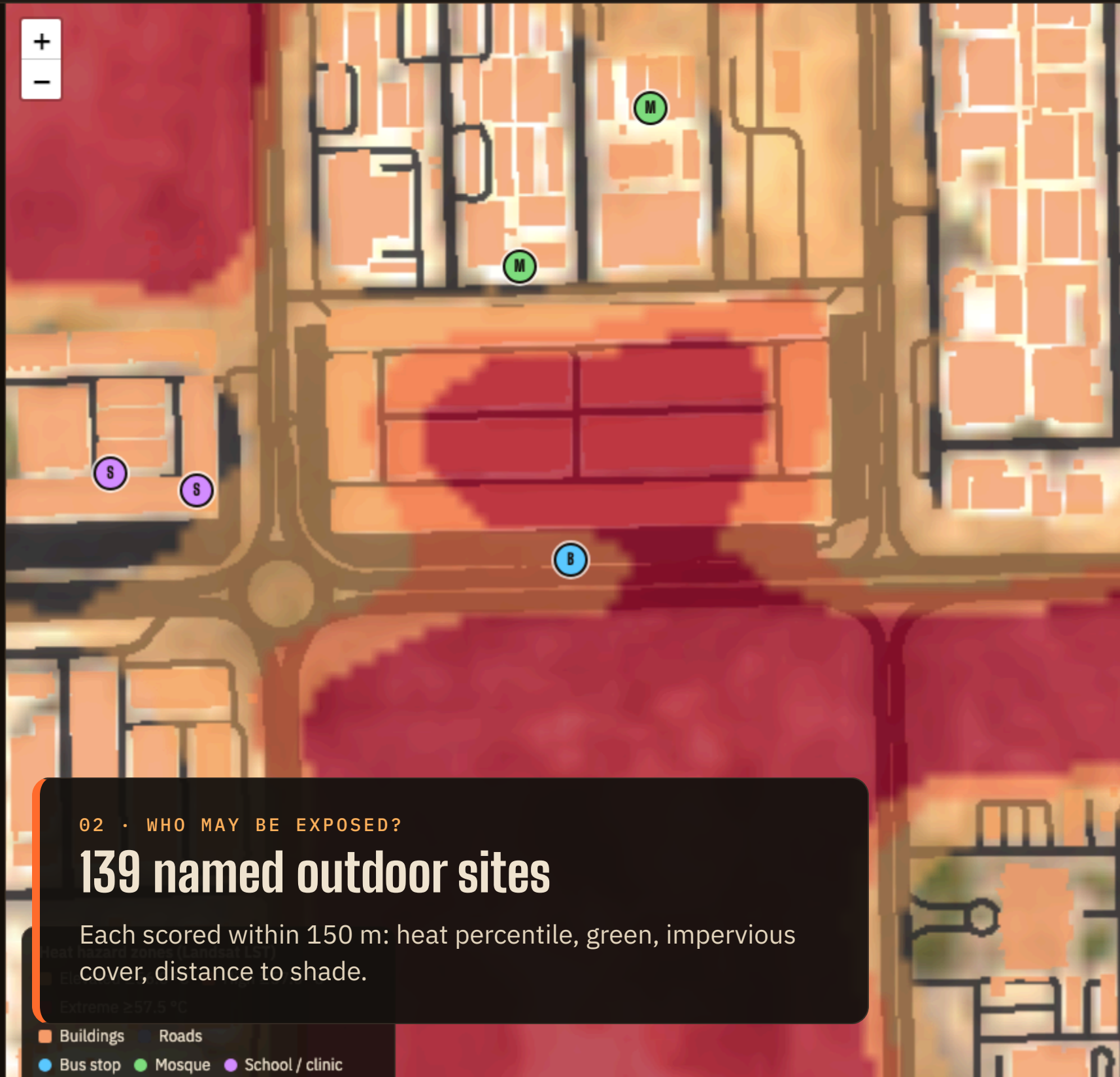
1.2 °C

mean error

05 · WHAT SHOULD BE DONE?

Where to act first in Musaffah

Ranked 100 m cells with the action, the reason in plain words, and the relative potential of trees



02 · WHO MAY BE EXPOSED?

139 named outdoor sites

Each scored within 150 m: heat percentile, green, impervious cover, distance to shade.

- Buildings
- Roads
- Bus stop
- Mosque
- School / clinic

VERY HIGH

BUS STOP

(unnamed)

HEAT	P97 within 150 m
GREEN	0% vegetation · nearest green 230 m
SURFACE	64% impervious
CONTEXT	public transport

RECOMMENDED

Shaded / cooled bus shelter + cool pavement

Hi-res imagery

Satellite

Buildings & roads

Heat hazard

Surfaces

Confidence

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1.2 °C

mean error

05 · WHAT SHOULD BE DONE?

Where to act first in Musaffah

Ranked 100 m cells with the action, the reason in plain words, and the relative potential of trees, cool pavement and shaded stops.

M-001 High

top hotspot

priority

1

M-001 · industrial outdoor workers
Shade structures

57.2°

2

M-002 · delivery riders
Shade structures

57.6°



03 · WHAT IS PHYSICALLY THERE?

Road · roof · sand · green · water

Held-out blocks: macro-F1 0.91. Starter index rules: 0.47.

Surfaces (Sentinel-2, 10 m)

■ Road / dark pavement

■ Building / roof

■ Bare soil / sand

■ Vegetation

■ Water

Musaffah deep dive

Abu Dhabi · 9 × 9 km at 10 m

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Hi-res imagery

Satellite

Buildings & roads

Heat hazard

Surfaces

Confidence

People places

Exposure sites

Priority

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0.85 **1.2 °C**

R², spatial CV mean error

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M-001 **High**

top hotspot priority

- | | | |
|---|--|-------|
| 1 | M-001 · industrial outdoor workers
Shade structures | 57.2° |
| 2 | M-002 · delivery riders
Shade structures | 57.6° |
| 3 | M-003 · industrial outdoor workers
Shade structures | 57.3° |
| 4 | M-004 · delivery riders
Shade structures | 56.8° |
| 5 | M-005 · delivery riders
Shade structures | 57.6° |
| 6 | M-006 · delivery riders
Shade structures | 57.7° |



04 · WHY MAY IT BE HOT?

Asphalt and sand up, green down

Spatially validated: R² 0.85, error 1.2 °C. Shown as associations, not proof of cause.

Heat hazard zones (estimated °C)

■ Elevated ≥56.0 °C ■ High ≥57.0 °C

■ Extreme ≥57.5 °C

■ Buildings ■ Roads

Musaffah deep dive

Abu Dhabi · 9 × 9 km at 10 m

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M-001 High

top hotspot

priority

- 1

M-001 · industrial outdoor workers

Shade structures

57.2°
- 2

M-002 · delivery riders

Shade structures

57.6°
- 3

M-003 · industrial outdoor workers

Shade structures

57.3°
- 4

M-004 · delivery riders

Shade structures

56.8°
- 5

M-005 · delivery riders

Shade structures

57.6°
- 6

M-006 · delivery riders

Shade structures

57.7°
- 7

M-007 · industrial outdoor workers

Shade structures

57.9°
- 8

M-008 · industrial outdoor workers

Shade structures

56.7°
- 9

M-009 · industrial outdoor workers

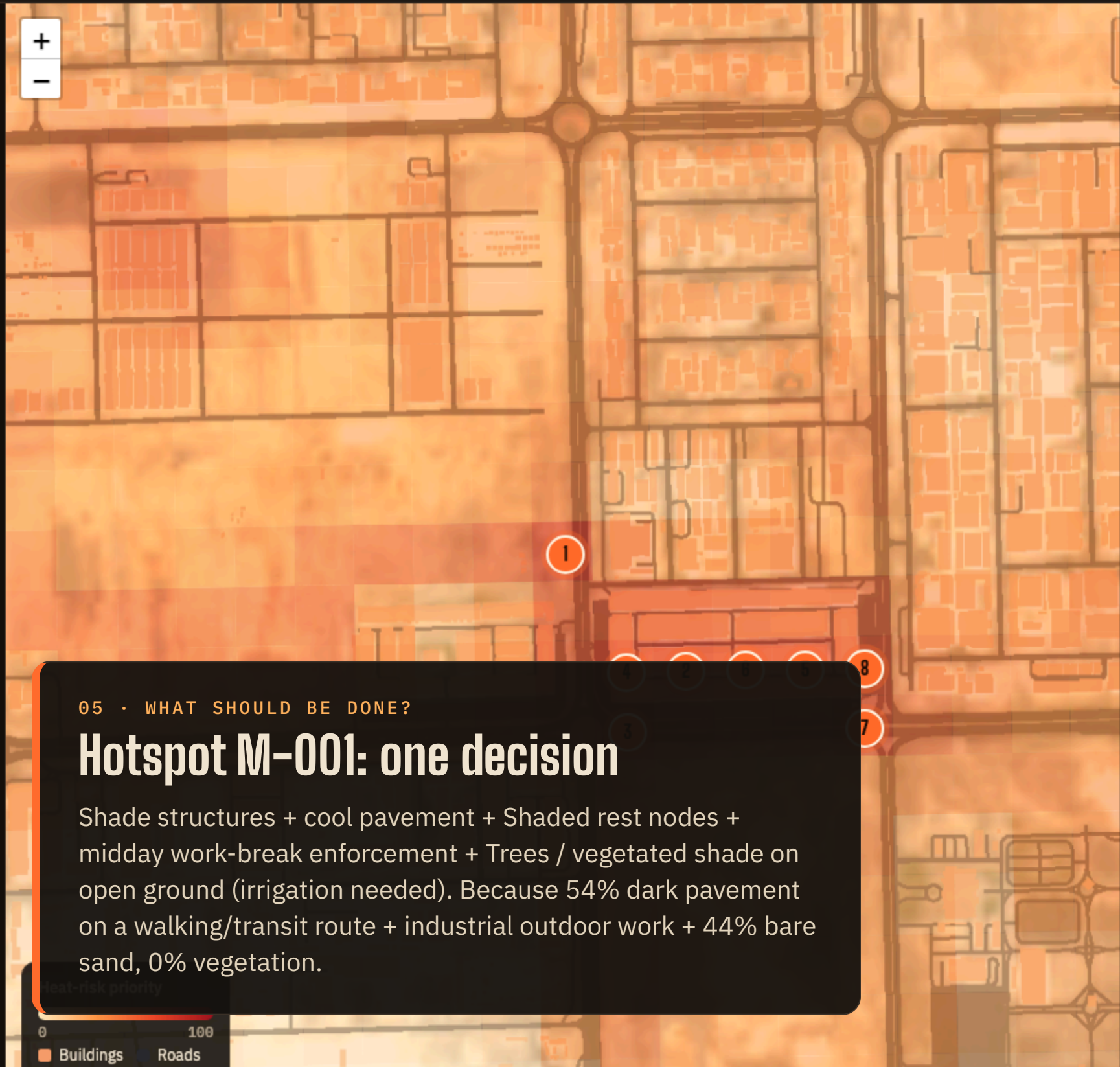
Shade structures

56.3°
- 10

M-010 · delivery riders

Shade structures

57.2°



05 · WHAT SHOULD BE DONE?

Hotspot M-001: one decision

Shade structures + cool pavement + Shaded rest nodes + midday work-break enforcement + Trees / vegetated shade on open ground (irrigation needed). Because 54% dark pavement on a walking/transit route + industrial outdoor work + 44% bare sand, 0% vegetation.

HIGH

HIGH CONFIDENCE

RISK 82

Musaffah hotspot M-001

WHERE

24.3486, 54.5119 · [open map ↗](#)

HEAT

P93 thermal hazard · 57.2 °C zone

WHO

industrial outdoor workers, pedestrians · 24 residents

WHAT

road 54% · roof 2% · sand 44% · green 0.0% · nearest green 230 m

WHY

asphalt / dark pavement ↑, bare sand ↑, surface brightness (albedo) ↓

WHAT SHOULD BE DONE

- Shade structures
- cool pavement
- Shaded rest nodes
- midday work-break enforcement
- Trees / vegetated shade on open ground (irrigation needed)

Because: 54% dark pavement on a walking/transit route + industrial outdoor work + 44% bare sand, 0% vegetation



04 · WHY

Roof, road or sand?
Tanager reads the surface

426-band Tanager spectra, labelled automatically from OpenStreetMap + Microsoft building outlines and WorldCover, sort every pixel into road, roof, vegetation and sand. Tested on tiles the model never saw.

0.76

full spectrum F1

0.75

6 broad bands

0.38

starter index rules

05 · WHO

Who is outside in it

Mosques, bus stops, delivery-rider hubs, construction sites, schools and clinics on WorldPop residents. The hottest fifth of neighbourhoods has 0.00 mapped bus stops per 10,000 people, against 0.63 elsewhere.

157,076

residents in hottest 20%

90%

priorities changed by counting people

06 · WHAT NEXT

Where to act first

300 m cells ranked by heat × people × missing greenery. Pick a hotspot to see who is exposed, why it is hot, what to do and the estimated cooling.

-0.70 °C

per +0.10 roof albedo

+0.13

R² gained from hyperspectral



06 · HYPERSPECTRAL PROOF · RIYADH

Tanager reads roof, road and sand

Full spectrum F1 0.76 vs 6 bands 0.75 on unseen tiles, and +0.13 R² for heat.

Surface (Tanager)

Road / asphalt

Roof / building

Vegetation

Bare / sand

Water

East Riyadh

As Sali · Saudi Arabia

WHERE

Persistent summer hotspots at 300 m

WHEN

Hours above 40 °C, from weather reanalysis

WHY

Roof, road, sand or missing greenery, read from hyperspectral spectra

WHO

Residents + people outdoors: mosque-goers, bus riders, delivery riders, workers, children

WHAT NEXT

One named action per hotspot, with estimated °C and confidence

Open the last chapter, “Where to act first”, and pick a hotspot.

03 · WHY

The starter rule calls sand a city

NDBI > 0 labels Musaffah 92% and Masdar 80% built-up. WorldCover says 70% and 17%. QAYDH's model: 82% and 35%.

80%

NDBI says built (Masdar)

17%

WorldCover

04 · WHO

Musaffah: workers, riders, worshippers

Musaffah holds 87,463 residents, 25 bus stops and 36 mosques in one hot industrial grid, the highest people-priority of the four districts.

53.7

Musaffah priority

43.1

Masdar priority

05 · WHAT NEXT

Where to act first in Abu Dhabi

Ranked 300 m cells with the action each one needs. Hyperspectral roof/road detail plugs in when Satellite 813 or MBZ-SAT data is available.

560,471

residents covered

1709

mapped outdoor places

1

156 residents

Cool roofs & cool pavements

56.6°

2

Musaffah industrial 191 residents

Shade trees / green corridors

56.7°

3

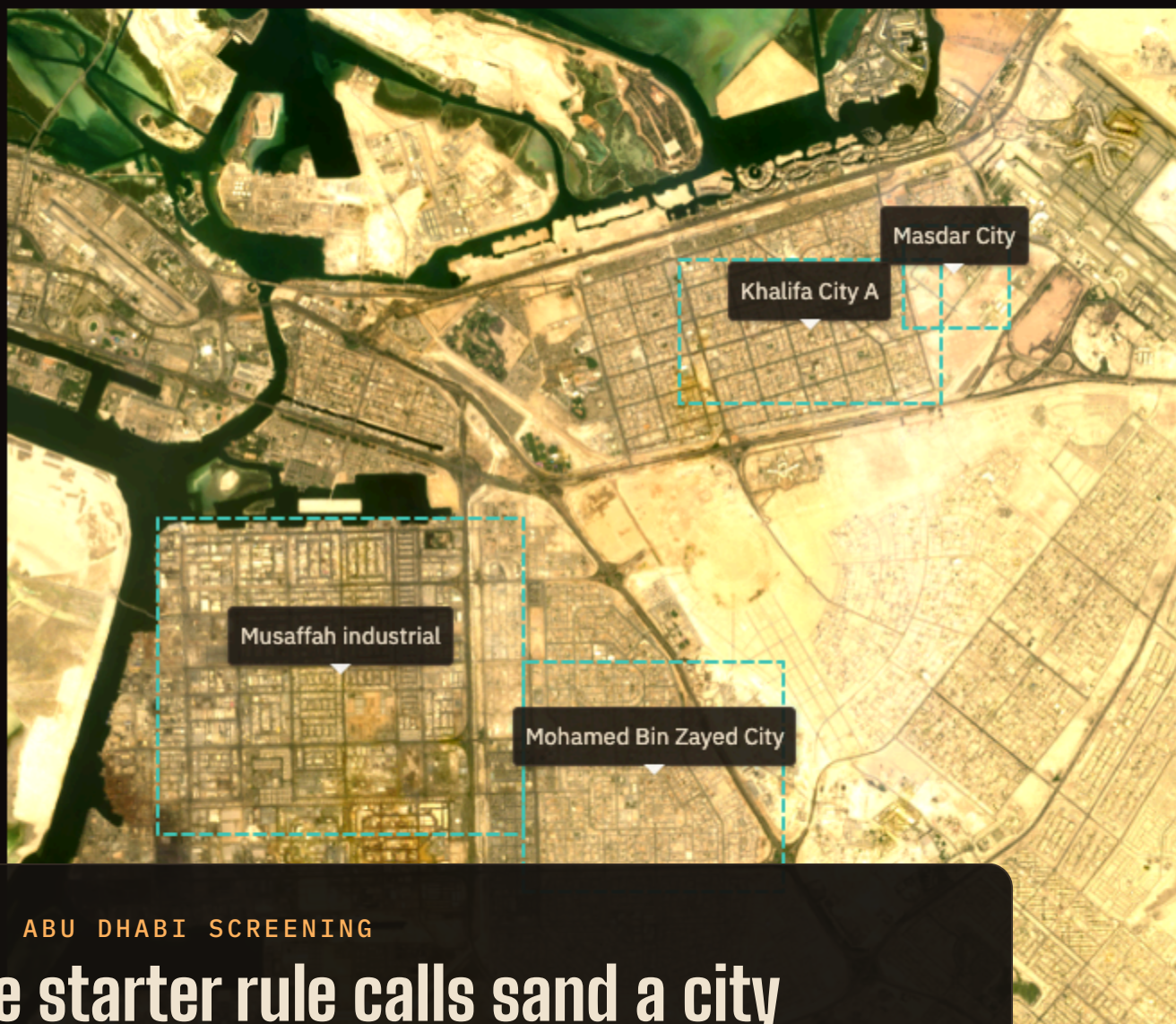
Musaffah industrial 183 residents

Shade trees / green corridors

56.3°

+

-



07 · ABU DHABI SCREENING

The starter rule calls sand a city

NDBI says Masdar 80% built; WorldCover 17%. Musaffah: 92% vs 70%.

Landsat 2025

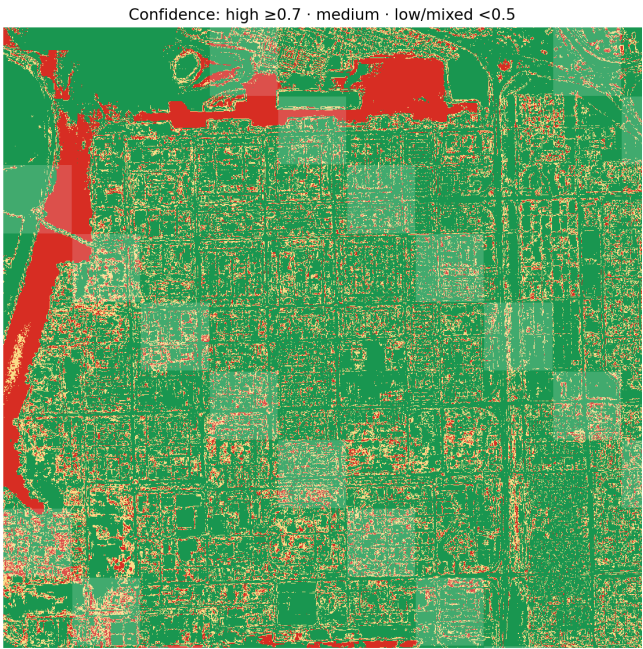
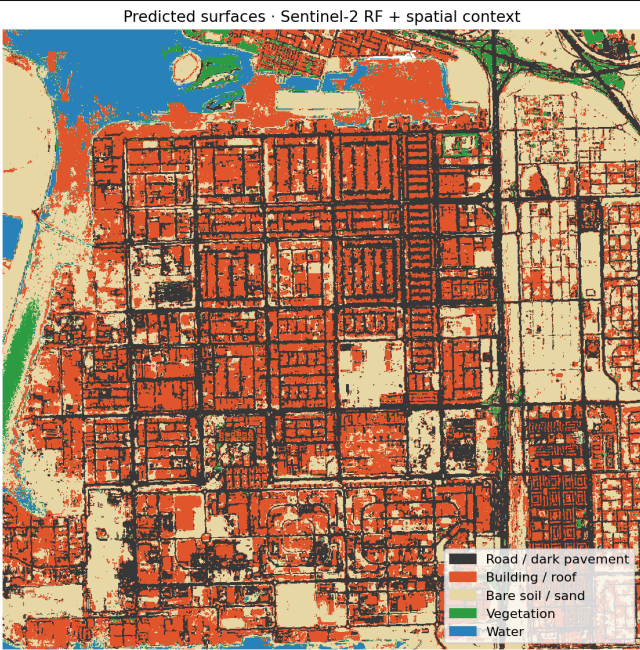
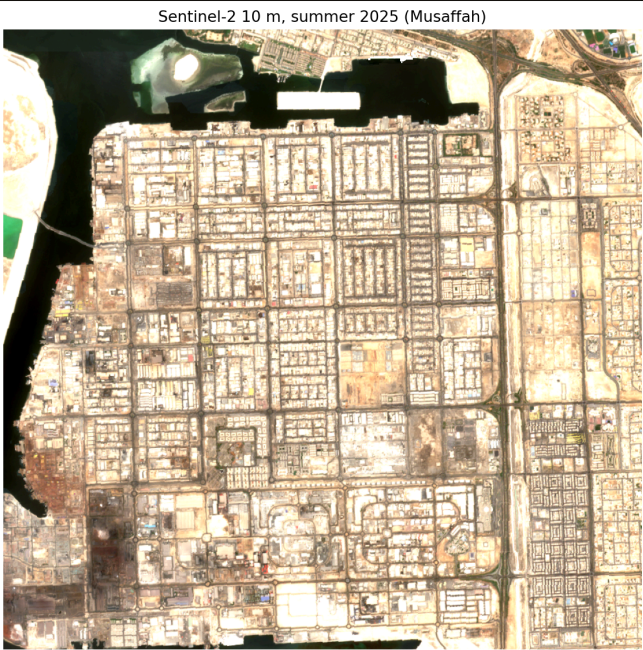
Abu Dhabi

Musaffah · Masdar · MBZ City · Khalifa City

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WHO	Residents + people outdoors: mosque-goers, bus riders, delivery riders, workers, children
WHAT NEXT	One named action per hotspot, with estimated °C and confidence

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Expert labels, honest tests



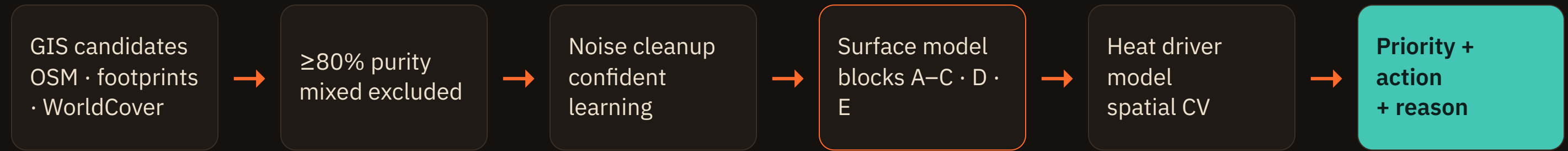
Index rules (starter NDVI/NDBI/BUI/MNDWI)
test macro-F1 0.47 (held-out blocks E)

Road	0.00	0.95	0.04	0.01	0.00
Building	0.00	0.51	0.37	0.00	0.12
Bare soil	0.00	0.92	0.07	0.00	0.01
Vegetation	0.00	0.00	0.00	0.99	0.01
Water	0.00	0.00	0.00	0.00	1.00
Road		Building	Bare soil	Vegetation	Water

Sentinel-2 RF + spatial context
test macro-F1 0.91 (held-out blocks E)

Road	0.87	0.05	0.08	0.00	0.00
Building	0.04	0.86	0.09	0.00	0.00
Bare soil	0.06	0.11	0.83	0.00	0.01
Vegetation	0.00	0.00	0.00	1.00	0.00
Water	0.00	0.01	0.00	0.02	0.98
Road		Building	Bare soil	Vegetation	Water

From labels to a decision



Supervised

Random Forest, pixel + context features

Validated

1 km blocks never seen in training

Explainable

Every action states its reason

WHY IT FITS

Matched to a desert city

Sand looks built

Index rules call 0% of roads right. The model: 88%.

Pixels are mixed

Only $\geq 80\%$ pure pixels train; confidence is mapped.

Neighbours leak

Train, validate, test on separate 1 km blocks.

Heat is coarse

Thermal = zones; 10 m surfaces explain them.

KNOWN TRADEOFFS

Being upfront

LABELS

Automatic candidates still carry noise. 1,718 removed; a 300-point review queue goes to QGIS.

HEAT

Surface, ~10:40. Air peaks later; ERA5 sets the hours.

MATERIALS

No open hyperspectral over Musaffah yet. 813 plugs in.

PEOPLE

Exposure opportunity, not head counts.

PROOF

Every claim has a number

0.91

Musaffah surfaces, held-out
starter rules 0.47

0.85

heat driver model R^2
spatial CV

0.89

built-up F1 Riyadh
starter 0.58

+0.13

R^2 from hyperspectral
Riyadh built-up

0.90

hotspots repeat
2024 ↔ 2025

60–100%

priorities stable
under re-weighting

Who benefits

Municipalities & urban planners

Abu Dhabi DMT • Dubai Municipality

Industrial-zone operators

Transport authorities (bus shelters)

Delivery platforms & contractors

Developers & master-planners

Solution value

PEOPLE

Protects people outdoors at the hours that matter

BUDGETS

Ranks shade, cool surfaces and trees by need

SCALE

Open data, one notebook, any Gulf city

ALIGNMENT

UAE Net Zero 2050 · We the UAE 2031
· SDG 3 · 11 · 13

QAYDH القفّظ

From “where is it hot?” to “where we act first, why, and how.”

Musaffah → Abu Dhabi → UAE → Gulf cities